Rao Test for Radar Target Detection in Sub-Gaussian Symmetric Alpha-Stable Sea Clutter

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***Abstract*—This paper presents an adaptive detector based on maximum a posteriori (MAP) Rao test specially for the sub-Gaussian symmetric alpha-stable (SGSαS) sea clutter background. Since SGSαS distribution does not have a closed-form probability density function (PDF) expression, there are few researches on the problem of target detection in SGSαS clutter background. In this paper, firstly the closed-form PDF expression of the SGSαS distribution is given in terms of the Fox's H-function. Then the MAP Rao-SGSαS detector in alpha-stable clutter is designed based on two-step MAP Rao test. Experimental results based on the simulated data verify the effectiveness of the proposed detector.**

Keywords—Sub-Gaussian symmetric alpha-stable (SGSαS) distribution, Rao test, Alpha-stable distribution, Coherent detection

I. Introduction

In the modern battlefield, the radar echo signal inevitably contains the sea clutter when detecting the sea surface target or low altitude targets over sea. A large number of researches show that radar clutter not only has significant non-Gaussian characteristics, but also has complex correlation characteristics [1], [2]. Especially for a high-resolution radar system and/or in the case of a low grazing angle, the sea clutter show heavy-tailed statistical characteristics.

To model the non-Gaussian sea clutter, some statistic models are proposed. Reasoning on the physical radar scenario, the class of spherically invariant random process (SIRP) is proposed to model radar sea clutter process [3]. SIRP model can be seen as a fast varying speckle component modulated by a slowly varying texture component. When the texture of SIRP is model with different distributions, the clutter amplitude or power follows different distributions. The clutter amplitude becomes the K distribution when the texture component is gamma distributed; the clutter power becomes the generalized Pareto distribution when the texture component is inverse-gamma distributed; and the clutter amplitude becomes the compound-Gaussian with inverse Gaussian texture (CG-IG) distribution when the texture component is inverse-Gaussian distributed.

Alpha-stable distribution is another family of distributions used to model the non-Gaussian sea clutter especially the heavy-tailed sea clutter. Alpha-stable distribution can be seen as a generalized Gaussian distribution and it has some significant properties. Firstly, the limiting distribution of the sum of the IID random variables follows the alpha-stable distribution. In other words, the alpha-stable distribution satisfies the generalized central limit theorem. Secondly, the sum of different independent alpha-stable random variables with the same characteristic exponent α is also an alpha-stable variable. Besides, the alpha-stable distribution is suitable to model the heavy-tailed sea clutter since the probability density function (PDF) of alpha-stable distribution decays algebraically at the tail. For the radar target coherent detection, a special vector form of alpha-stable distribution, the sub-Gaussian symmetric alpha-stable (SGSαS) process, is proposed to model the clutter sample vectors [4], [5]. However, since the SGSαS distribution generally does not have a closed-form PDF expression, the research on detector design in SGSαS clutter background is limited.

In radar target detection, the optimal detectors are always derived from the Neyman-Pearson criterion. However, there are always some unknown parameters in practice so that the optimal detectors cannot be derived. The generalized likelihood ratio test (GLRT) is proposed as a sub-optimal detectors and the unknown parameters in GLRT need to be estimated. Besides, there are other two sub-optimal detectors for practical application, which are the Rao and Wald test. In [6], it is pointed out that the Rao and Wald test have more robust performance than the GLRT and they have less computational complexity than the GLRT. In this paper, a two-step detector based on maximum a posteriori (MAP) Rao test in SGSαS clutter background is derived. Firstly, the PDF of SGSαS distribution is derived based on the Fox's H-function. Secondly, based on the two-step MAP Rao test, the MAP Rao-SGSαS detector is derived.

The remainder of the paper is organized as follows. In Section II, the Fox's H-function is introduced firstly and the PDF of the SGSαS distribution is derived. In Section III, the detection problem is formulated and the MAP Rao-SGSαS detector is derived. In Section IV, some results and

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discussions are given. Finally, some conclusions are given in Section V.

II. The PDF of the Sub-Gaussian Symmetric Alpha-Stable Clutter

A. The Fox's H-function

The Fox's H-function [7], which is known the H-function for short, is defined in terms of a Mellin-Barnes integral:

$$\begin{aligned} H_{p,q}^{m,n}\left[z \left| \begin{matrix} (a_i, A_i)_{1,p} \\ (b_j, B_i)_{1,q} \end{matrix} \right. \right] &= H_{p,q}^{m,n}\left[z \left| \begin{matrix} (a_1, A_1), \cdots, (a_p, A_p) \\ (b_1, B_1), \cdots, (b_q, B_q) \end{matrix} \right. \right] \\ &= \frac{1}{2\pi i} \int_{\mathcal{L}} \Theta(s) x^{-s} ds \end{aligned} \tag{1}$$

with

$$\Theta(s) = \frac{\prod_{j=1}^{m} \Gamma(b_j + B_j s) \prod_{i=1}^{n} \Gamma(1 - a_i - A_i s)}{\prod_{i=n+1}^{p} \Gamma(a_i + A_i s) \prod_{j=m+1}^{q} \Gamma(1 - b_j - B_j s)} \tag{2}$$

where m, n, p, q are integers with $0 \le m \le q$ and $0 \le n \le p$; z is a complex variable with $z \ne 0$; a_i and b_j are complex numbers; A_i and B_j are positive real numbers; $\mathcal{L}$ is the integral path, which should be chosen to separate the poles of $\Gamma(b_j + B_j s)$ from the poles of $\Gamma(1 - a_i - A_i s)$.

B. The Sub-Gaussian Symmetric Alpha-stable Distribution

In this paper, the SGSαS is used to model the complex clutter vector $\mathbf{c}$. According to the sub-Gaussian Theory [4], an SGSαS variable $\mathbf{c}$ can be written as:

$$\mathbf{c} = \sqrt{s}\mathbf{g} \tag{3}$$

where s is a positive alpha-stable (PαS) variable, $\mathbf{g}$ is a circular complex Gaussian vector with zero mean and covariance matrix $\mathbf{R}$. From (3), we can see that the SGSαS distribution model is a kind of the SIRP model with a texture component that obeys the PαS distribution.

Since the PαS distribution generally does not have a closed-form PDF expression of elementary function, it is always expressed by its characteristic function (CF). Then the CF of the PαS texture s is in the following form [8]:

$$\varphi(t) = \exp(-\gamma t^{\alpha}) \tag{4}$$

where α is the characteristic exponent reflecting the shape of PαS distribution; γ is the scale parameter reflecting the spread of the PαS distribution with $\gamma > 0$.

In [8], in order to analyze the detectors in PαS clutter background, the closed form PDF of texture component s in terms of the H-function is given:

$$p_s(s) = s^{-1} \mathrm{H}_{1,1}^{0,1}\left[\gamma^{-1} s^{\alpha} \left| \begin{matrix} (1,1) \\ (1,\alpha) \end{matrix} \right. \right] = s^{-1} \mathrm{H}_1\left[\gamma^{-1} s^{\alpha} \right] \tag{5}$$

where the $\mathrm{H}_1[\bullet]$ is used to express the $\mathrm{H}_{1,1}^{0,1}[\bullet|\bullet]$ for simplicity. Then according to (3), the PDF $p_{\mathbf{c}}(\mathbf{c})$ of the SGSαS vector $\mathbf{c}$ can be easily written as:

$$\begin{aligned} p_{\mathbf{c}}(\mathbf{c}) &= \int_0^{\infty} p_{\mathbf{c}|s}(\mathbf{c} \mid s) p_s(s) ds \\ &= \int_0^{\infty} \frac{1}{(\pi s)^N |\mathbf{R}|} \exp\left(-\frac{\mathbf{c}^H \mathbf{R}^{-1} \mathbf{c}}{s} \right) p_s(s) ds \end{aligned} \tag{6}$$

where $p_{\mathbf{c}|s}(\mathbf{c} \mid s)$ is the conditional PDF for the random vector $\mathbf{c}$ given the PαS variable s and $p_s(s)$ is the PDF of s. Based on the (2.9.4) in [7], the exponential function in (6) can be written as other form of the H-function:

$$\exp\left(-\frac{\mathbf{c}^H \mathbf{R}^{-1} \mathbf{c}}{s} \right) = \mathrm{H}_{1,1}^{1,1}\left[\frac{s}{\mathbf{c}^H \mathbf{R}^{-1} \mathbf{c}} \left| \begin{matrix} (1,1) \\ (1,0) \end{matrix} \right. \right] \tag{7}$$

Substituting (7) to (6) and using the integral theorem in [7], (6) becomes:

$$\begin{aligned} p_{\mathbf{c}}(\mathbf{c}) &= \frac{\pi^{-N}}{|\mathbf{R}|} (\mathbf{c}^H \mathbf{R}^{-1} \mathbf{c})^{-N} \mathrm{H}_{1,2}^{1,1}\left[\gamma^{-1} (\mathbf{c}^H \mathbf{R}^{-1} \mathbf{c})^{\alpha} \left| \begin{matrix} (1,1) \\ (\mathrm{N},\alpha)(1,\alpha) \end{matrix} \right. \right] \\ &\equiv \frac{\pi^{-N}}{|\mathbf{R}|} (\mathbf{c}^H \mathbf{R}^{-1} \mathbf{c})^{-N} \mathrm{H}_2\left[\gamma^{-1} (\mathbf{c}^H \mathbf{R}^{-1} \mathbf{c})^{\alpha} \right] \end{aligned} \tag{8}$$

where the $\mathrm{H}_2[\bullet]$ is used to express the $\mathrm{H}_{1,2}^{1,1}[\bullet|\bullet]$ for simplicity.

III. Detection Problem Formulation and Rao Test Scheme

In this section, firstly the coherent detection problem in SGSαS clutter background is formulated. Then the two-step scheme is adopted to design the Rao test for the coherent detection problem in SGSαS clutter background.

A. Detection Problem Formulation

A detection problem is indeed a binary hypothesis test problem. Here we consider a clutter-dominant background where the noise can be ignored and consider the received signal $\mathbf{z}$ is a $N \times 1$ column vector. Then the received signal $\mathbf{z}$ only consists of clutter $\mathbf{c}$ under hypothesis H_0 (target is absent) and $\mathbf{z}$ consists of clutter vector $\mathbf{c}$ and target signal vector $\mathbf{s}$ under hypothesis H_1 (target is present):

$$\begin{cases} H_0 : \mathbf{z} = \mathbf{c} \\ H_1 : \mathbf{z} = \beta \mathbf{p} + \mathbf{c} \end{cases} \tag{9}$$

where the $\beta \mathbf{p}$ represents the target signal, β is the target signal amplitude, and $\mathbf{p}$ is the steering vector. Here $\mathbf{p}$ is assumed known, β is assumed as an unknown but deterministic parameter.

B. Rao Test Scheme

Here the two-step scheme is adopted to design the Rao test for the coherent detection problem in SGSαS clutter background. Specifically, the PαS variable s and the covariance matrix $\mathbf{R}$ are assumed known and the Rao test is derived firstly. Then the maximum MAP of s and the fixed point (FP) estimate of $\mathbf{R}$ are substituted in the Rao test derived in the first step, and the MAP Rao-SGSαS detector can be obtained. In order to obtain the Rao test statistic, the notations are defined firstly:

- $\boldsymbol{\theta}_r = [\beta_R, \beta_I]^T$ is a two-dimensional real vector, where the β_R and β_I are the real and imaginary parts of the parameter β.
- $\boldsymbol{\theta}_s = s$ is the nuisance parameter.
- $\boldsymbol{\theta} = \left[\boldsymbol{\theta}_r^T, \boldsymbol{\theta}_s^T\right]^T$ is the three-dimensional real vector.

1) Step 1

In Step 1, we assume that s and $\mathbf{R}$ are all known. The Rao test [9] for the problem shown in (9) is:

$$\frac{\partial \ln p(\mathbf{z}\,|\,\beta,\mathbf{R},s;H_1)}{\partial \boldsymbol{\theta}_r}\bigg|^T_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}_0}\left[\mathbf{J}^{-1}\left(\hat{\boldsymbol{\theta}}_0\right)\right]_{\boldsymbol{\theta}_r,\boldsymbol{\theta}_r} \times \frac{\partial \ln p(\mathbf{z}\,|\,\beta,\mathbf{R},s;H_1)}{\partial \boldsymbol{\theta}_r}\bigg|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}_0} \underset{H_0}{\overset{H_1}{\gtrless}} T_{Rao} \tag{10}$$

where:

- T_{Rao} is the detection threshold;
- $\hat{\boldsymbol{\theta}}_0 = \left[\hat{\boldsymbol{\theta}}_{r,0}^T, \hat{\boldsymbol{\theta}}_s^T\right]^T$ is the maximum likelihood (ML) estimate of parameter vector $\boldsymbol{\theta}$ under hypothesis H_0. Since the parameter $\boldsymbol{\theta}_s$ is assumed known in the first step, here only the ML estimate $\hat{\boldsymbol{\theta}}_{r,0}$ of $\boldsymbol{\theta}_r$ needs to be derived.
- $p(\mathbf{z}\,|\,\beta,\mathbf{R},s;H_1)$ is the conditional PDF for $\mathbf{c}$ given β, $\mathbf{R}$ and s under hypothesis H_1, $\mathbf{J}(\boldsymbol{\theta})$ is the Fisher information matrix, which is donated as follows:

$$\mathbf{J}(\boldsymbol{\theta}) = \begin{bmatrix} \mathbf{J}_{\boldsymbol{\theta}_r,\boldsymbol{\theta}_r}(\boldsymbol{\theta}) & \mathbf{J}_{\boldsymbol{\theta}_r,\boldsymbol{\theta}_s}(\boldsymbol{\theta}) \\ \mathbf{J}_{\boldsymbol{\theta}_s,\boldsymbol{\theta}_r}(\boldsymbol{\theta}) & \mathbf{J}_{\boldsymbol{\theta}_s,\boldsymbol{\theta}_s}(\boldsymbol{\theta}) \end{bmatrix} \tag{11}$$

and

$$\left[\mathbf{J}^{-1}(\boldsymbol{\theta})\right]_{\boldsymbol{\theta}_r,\boldsymbol{\theta}_r} = \left[\mathbf{J}_{\boldsymbol{\theta}_r,\boldsymbol{\theta}_r}(\boldsymbol{\theta}) \text{-} \mathbf{J}_{\boldsymbol{\theta}_r,\boldsymbol{\theta}_s}(\boldsymbol{\theta})\mathbf{J}^{-1}_{\boldsymbol{\theta}_s,\boldsymbol{\theta}_s}(\boldsymbol{\theta})\mathbf{J}_{\boldsymbol{\theta}_s,\boldsymbol{\theta}_r}(\boldsymbol{\theta})\right]^{-1} \tag{12}$$

It is easy to obtain that:

$$\frac{\partial \ln p(\mathbf{z}\,|\,\beta,\mathbf{R},s;H_1)}{\partial \beta_R} = 2\,\mathrm{Re}\left\{\frac{\mathbf{p}^H\mathbf{R}^{-1}(\mathbf{z}-\beta\mathbf{p})}{s}\right\} \tag{13}$$

$$\frac{\partial \ln p(\mathbf{z}\,|\,\beta,\mathbf{R},s;H_1)}{\partial \beta_I} = 2\,\mathrm{Im}\left\{\frac{\mathbf{p}^H\mathbf{R}^{-1}(\mathbf{z}-\beta\mathbf{p})}{s}\right\} \tag{14}$$

where $\mathrm{Re}\{\bullet\}$ and $\mathrm{Im}\{\bullet\}$ represent the real and imaginary parts. Then it can be obtained easily that:

$$\frac{\partial \ln p(\mathbf{z}\,|\,\beta,\mathbf{R},s;H_1)}{\partial \boldsymbol{\theta}_r} = 2\left[\mathrm{Re}\left\{\frac{\mathbf{p}^H\mathbf{R}^{-1}(\mathbf{z}-\beta\mathbf{p})}{s}\right\}, \mathrm{Im}\left\{\frac{\mathbf{p}^H\mathbf{R}^{-1}(\mathbf{z}-\beta\mathbf{p})}{s}\right\}\right]^T \tag{15}$$

Then the blocks $\mathbf{J}_{\boldsymbol{\theta}_r,\boldsymbol{\theta}_r}(\boldsymbol{\theta})$ and $\mathbf{J}_{\boldsymbol{\theta}_r,\boldsymbol{\theta}_s}(\boldsymbol{\theta})$ of the Fisher information matrix can be derived as:

$$\mathbf{J}_{\boldsymbol{\theta}_r,\boldsymbol{\theta}_r}(\boldsymbol{\theta}) = \begin{bmatrix} \dfrac{\mathbf{p}^H\mathbf{R}^{-1}\mathbf{p}}{s} & 0 \\ 0 & \dfrac{\mathbf{p}^H\mathbf{R}^{-1}\mathbf{p}}{s} \end{bmatrix} \tag{16}$$

$$\mathbf{J}_{\boldsymbol{\theta}_r,\boldsymbol{\theta}_s}(\boldsymbol{\theta}) = [0,0]^T \tag{17}$$

According to (12), (16) and (17), the $\left[\mathbf{J}^{-1}(\boldsymbol{\theta})\right]_{\boldsymbol{\theta}_r,\boldsymbol{\theta}_r}$ in (10) can be obtained:

$$\left[\mathbf{J}^{-1}(\boldsymbol{\theta})\right]_{\boldsymbol{\theta}_r,\boldsymbol{\theta}_r} = \begin{bmatrix} \dfrac{s}{2\mathbf{p}^H\mathbf{R}^{-1}\mathbf{p}} & 0 \\ 0 & \dfrac{s}{2\mathbf{p}^H\mathbf{R}^{-1}\mathbf{p}} \end{bmatrix} \tag{18}$$

Finally substitute $\hat{\boldsymbol{\theta}}_{r,0} = [0,0]^T$, (15) and (18) into (10), the Rao test expression can be derived as:

$$\frac{\left|\mathbf{p}^H\mathbf{R}^{-1}\mathbf{z}\right|^2}{s\mathbf{p}^H\mathbf{R}^{-1}\mathbf{p}} \underset{H_0}{\overset{H_1}{\gtrless}} T'_{Rao} \tag{19}$$

2) Step 2

In Step 2, the MAP estimate $\hat{s}_{\mathrm{MAP}}$ under hypothesis H_0 of s and the FP estimate $\hat{\mathbf{R}}_{FP}$ of covariance matrix $\mathbf{R}$ are obtained to replace s and $\mathbf{R}$ in (19).

According to [9], we can obtain that:

$$\begin{aligned} & p(s\,|\,\mathbf{z};H_0) \\ & \propto p(\mathbf{z}\,|\,s;H_0)\,p(s;H_0) \\ & = \frac{1}{\pi^N s^{N+1}|\mathbf{R}|}\exp\left(-\frac{\mathbf{z}^H\mathbf{R}^{-1}\mathbf{z}}{s}\right)\mathrm{H}_1\left[\gamma^{-1}s^{\alpha}\right] \end{aligned} \tag{20}$$

Setting that:

$$\frac{\partial \ln p(s\,|\,\mathbf{z};H_0)}{\partial \boldsymbol{\theta}_s} = 0 \tag{21}$$

then (21) can be derived as:

$$-\frac{N+1}{s} + \frac{\mathbf{z}^H\mathbf{R}^{-1}\mathbf{z}}{s^2} + \frac{\left\{\mathrm{H}_1\left[\gamma^{-1}s^{\alpha}\right]\right\}'}{\mathrm{H}_1\left[\gamma^{-1}s^{\alpha}\right]} = 0 \tag{22}$$

where $\left\{\mathrm{H}_1\left[\gamma^{-1}s^{\alpha}\right]\right\}'$ is the partial derivative of $\mathrm{H}_1\left[\gamma^{-1}s^{\alpha}\right]$ with respect to $\boldsymbol{\theta}_s$. According to the differentiation formulas in [7], the $\left\{\mathrm{H}_1\left[\gamma^{-1}s^{\alpha}\right]\right\}'$ can be derived as:

$$\begin{aligned} \left\{\mathrm{H}_1\left[\gamma^{-1}s^{\alpha}\right]\right\}' &= -\frac{1}{s}\mathrm{H}^{1,1}_{2,2}\left[\gamma^{-1}s^{\alpha}\,\middle|\,\begin{matrix}(1,1)(0,\alpha)\\(1,\alpha)(1,\alpha)\end{matrix}\right] \\ &\equiv -\frac{1}{s}\mathrm{H}_3\left[\gamma^{-1}s^{\alpha}\right] \end{aligned} \tag{23}$$

where $\mathrm{H}_3[\bullet]$ is used to express the $\mathrm{H}^{1,1}_{2,2}[\bullet|\bullet]$ for simplicity.

Substitute (23) into (22), (22) becomes:

$$-\frac{N+1}{s} + \frac{\mathbf{z}^H\mathbf{R}^{-1}\mathbf{z}}{s^2} - \frac{\mathrm{H}_3\left[\gamma^{-1}s^{\alpha}\right]}{s\,\mathrm{H}_1\left[\gamma^{-1}s^{\alpha}\right]} = 0 \tag{24}$$

By solve the equation in (24), the MAP estimate $\hat{s}_{\mathrm{MAP}}$ of s can be obtained.

The $\hat{\mathbf{R}}_{FP}$ can be solved by the following iterative process:

$$\hat{\mathbf{R}}_{FP}^{(j+1)} = \frac{N}{K}\sum_{k=1}^{K}\frac{\mathbf{z}_k\mathbf{z}_k^H}{\mathbf{z}_k^H\left(\hat{\mathbf{R}}_{FP}^{(j)}\right)^{-1}\mathbf{z}_k} \tag{25}$$

with

$$\hat{\mathbf{R}}_{FP}^{(j+1)} = \frac{N}{\mathrm{Tr}\left(\hat{\mathbf{R}}_{FP}^{(j+1)}\right)}\hat{\mathbf{R}}_{FP}^{(j+1)} \tag{26}$$

and initial value

$$\hat{\mathbf{R}}_{FP}^{(0)} = \frac{N}{K}\sum_{k=1}^{K}\frac{\mathbf{z}_k\mathbf{z}_k^H}{\mathbf{z}_k^H\mathbf{z}_k} \tag{27}$$

where K is the number of secondary vector used to estimate the covariance matrix.

Finally, by substituting $\hat{s}_{\mathrm{MAP}}$ and $\hat{\mathbf{R}}_{FP}$ into (19), we get the MAP Rao-SGSαS detector:

$$\frac{\left|\mathbf{p}^H\hat{\mathbf{R}}_{FP}^{-1}\mathbf{z}\right|^2}{\hat{s}_{\mathrm{MAP}}\mathbf{p}^H\hat{\mathbf{R}}_{FP}^{-1}\mathbf{p}} \underset{H_0}{\overset{H_1}{\gtrless}} T'_{Rao} \tag{28}$$

IV. Results and Discussions

Here some experimental results based on simulated data are given and analyzed. In experiments, the target and clutter samples are simulated according to their models mentioned in this paper. 10^8 Monte Carlo simulations are done so that the detection thresholds can be obtained. Since the traditional signal-to-clutter ratio (SCR) cannot be calculated in SGSαS clutter, here a generalized signal-to-clutter ratio (GSCR) is adopted:

$$\mathrm{GSCR} = 10\log\left[\left(\frac{\beta}{\gamma^{1/\alpha}}\right)^{\alpha}\right] \tag{29}$$

The parameters are set as follows: $N=16$, $K=32$, $\gamma=1$, $\mathbf{p}=1/\sqrt{N}\left[1,\exp\left(j2\pi f_d\right),\ldots,\exp\left(j2\pi\left(N-1\right)f_d\right)\right]^T$ with f_d=0.3 and

$$\left[\mathbf{R}\right]_{ij} = \rho^{|i-j|}, 1\le i,j\le N \tag{30}$$

where ρ is the one-lag correlation coefficient and set to 0.9.

For comparison, the GLRT-SGSαS detector [4] and the adaptive normalized matched filter (ANMF) which is suitable for the detection problem in heavy-tailed clutter background are adopted. Fig. 1 gives the probability of detection (Pd) versus GSCR comparison for α is set to 0.5, 0.7 and 0.9 respectively. From Fig. 1, we can see that the proposed MAP Rao-SGSαS detector performs almost the same as the GLRT-SGSαS detector and the MAP Rao-SGSαS with FP estimate performs a little better than GLRT-SGSαS with FP estimate when parameter α is relative small. When $\alpha=0.7$, the MAP Rao-SGSαS with FP estimate still performs better than GLRT-SGSαS with FP estimate. When $\alpha=0.9$, MAP Rao-SGSαS with FP estimate performs a little worse than the GLRT-SGSαS with FP estimate, but it still performs better than the ANMF.

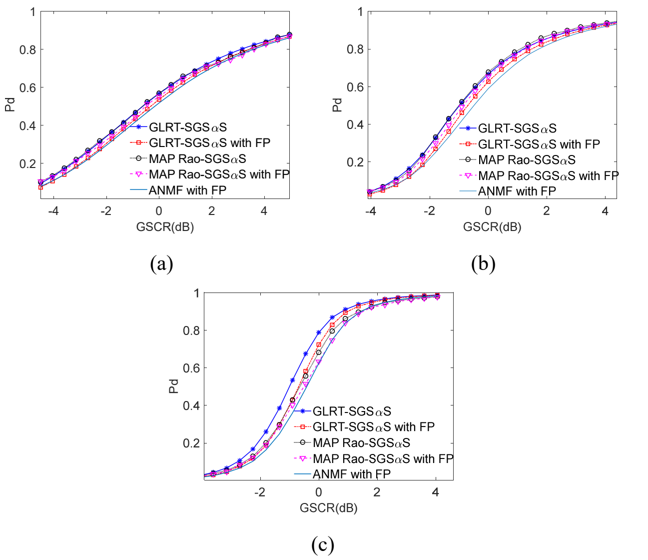

Fig. 1. Probability of detection versus GSCR on the simulated data. (*(a)* $\alpha=0.5$, *(b)* $\alpha=0.7$, *(c)* $\alpha=0.9$)

V. Conclusions

In this paper, we have designed the detector for radar targets embedded in the alpha-stable clutter background based on Rao test. Specifically, the clutter is modeled by the SGSαS distribution and the detector is designed based on the two-step MAP Rao test. Through the experiments based on the simulated data, the effectiveness of the proposed detector has been verified.

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